BARROW NFANSU O.

Embedded Systems and AI Engineering Student

CONTACT

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- Fes, Morocco

EDUCATION

2022 - 2027

ECOLE NATIONALE DES SCIENCES APPLIQUÉES DE FES

- Master's in Embedded Systems and AI

2018 - 2021

ST. PETER'S HIGH SCHOOL

- I studied Science as a field in The Gambia

2015 - 2018

ST. PETER'S JUNIOR SCHOOL

TECHNICAL SKILLS

- **Programming Languages:** C, C++, Python, Java, VHDL
- **Modeling & Simulation:** Matlab, Simulink, stateflow, MBD
- **Embedded Platforms:** Arduino, ESP32, Raspberry pi
- **Communication protocols:** UART, I2C, SPI, CAN, Ethernet, Lin
- **ML & AI:** Scikit-learn, Tensorflow, YOLO, openCV
- **Standards:** AUTOSAR, MISRA-C
- **Embedded Linux:** Buildroot
- **RTOS:** FreeRTOS
- **Development tools:** VS code, Linux, Arduino IDE, Thonny, Git
- **MBD:** V cycle

PROFILE

Embedded Systems and Artificial Intelligence engineering student at ENSAF, currently in the 2nd year of the engineering cycle, with practical experience in embedded software and microcontroller-based systems. **Currently seeking a PFA internship** to apply technical skills and gain practical industry experience.

PROJECTS

- **Custom Embedded Linux System – Raspberry Pi**

Developed a custom Embedded Linux image for Raspberry Pi using Buildroot, configuring and optimizing the system for embedded applications.
Technologies: Embedded Linux, Buildroot, Raspberry Pi

- **IoT-Based Remote Device Control – ESP32**

Designed and implemented an ESP32-based web server for remote control of electronic devices via Wi-Fi with real-time browser interaction.
Technologies: ESP32, Wi-Fi, Thonny, Sockets

- **Hand Gesture Controlled Embedded System**

Developed a computer-vision-based system for controlling electronic devices using hand gesture recognition integrated with ESP32.
Technologies: Python, OpenCV, ESP32, YOLO, mediapipe, Cvzone

- **Vehicle Detection and Counting System**

Implemented a computer vision system for detecting and counting vehicles from video streams using image-processing techniques.
Technologies: Python, OpenCV, YOLO, Cvzone.

- **Intelligent Vehicle Management Web Application – CHU**

Designed and developed a web-based vehicle management system during an internship at CHU to improve organization and tracking of hospital vehicle operations.
Technologies: HTML, CSS, JavaScript, YOLO, Flask, MYSQL, Bootstrap

SOFT SKILLS

- Team work
- Time management
- Continuous learning
- Communication Skills

CERTIFICATES

- Responsive Web Design Certification (Free Code Camp)
- Matlab Basic Certification (Math works)
- Intro to machine Learning
- Intro to deep learning (Kaggle)
- Computer Vision Certification (Kaggle)

LANGUAGES

- English (Fluent)
- French (Intermediate)
- Arabic (Basics)

EXPERIENCES

- Intern at CHU hospital at Fes.
- Treasurer of GASAM association in 2024/2025.